

Table 1: Accuracy (%) of investigated models and STORYREWARD on STORYRMB. *Average* represents the average score over the five dimensions. **Bold**: the best result. underline: the second-best result.

Model	Coherence	Creativity	Characterization	Fluency	Relevance	Average
InternLM2-20B-Reward	17.0	25.1	21.9	17.3	41.2	21.3
InternLM2-7B-Reward	18.5	18.9	18.1	15.6	25.0	18.7
Skywork-Reward-Gemma-2-27B-v0.2	36.6	29.8	22.8	40.0	34.6	32.7
Skywork-Reward-Llama-3.1-8B-v0.2	29.4	32.3	26.5	36.0	30.9	31.0
QRM-Llama3.1-8B-v2	15.1	17.9	9.3	13.3	19.9	14.9
ArmoRM-Llama3-8B-v0.1	52.4	48.8	44.2	56.4	43.4	49.7
GRM-Llama3.1-8B-rewardmodel-ft	35.1	28.1	23.7	35.1	35.3	31.1
WQRM	16.6	27.1	24.2	31.1	19.9	24.0
Llama3.1-8B-Instruct	22.3	30.5	25.1	16.9	24.9	24.3
Qwen3-8B	55.8	53.1	47.0	60.4	48.5	53.5
GPT-4o	63.4	<u>63.7</u>	<u>66.0</u>	<u>73.3</u>	67.6	<u>66.3</u>
Gemini-2.5-Pro	57.7	51.7	57.2	64.4	52.2	56.8
Grok-3	62.6	57.5	62.8	69.8	60.3	62.4
STORYREWARD-LLAMA	<u>64.5</u>	60.8	62.3	53.8	<u>67.7</u>	61.4
STORYREWARD-QWEN	77.5	78.8	71.6	74.2	69.1	75.0

is intuitive, since both reward models and LLM-as-a-judge are inherently sensitive to surface fluency (Tian et al., 2024). In contrast, for more complex dimensions requiring deeper narrative understanding, such as *creativity* and *characterization*, the scores are generally lower. This underscores the inability of existing reward models to capture story-specific preference and highlights the necessity of a dedicated reward model for stories. (3) Our trained models (STORYREWARD-QWEN and STORYREWARD-LLAMA) achieve the best results on STORYRMB, significantly outperforming the base models (Llama3.1 and Qwen3), particularly on story-specific metrics such as *creativity* and *characterization*. This demonstrates the effectiveness of our training data. As our data construction method is scalable, we also encourage the community to collect more high-quality corpora to develop more advanced reward models.

4.2 PREFERENCE ANALYSIS: HUMAN VS. LLM STORIES

Previous studies have shown that LLM-generated stories differ largely from human-authored ones (Tian et al., 2024), remaining far from human standards. We argue that a key reason is that existing reward models often favor LLM-generated stories. To investigate this, we analyze current reward models. As noted in § 2, STORYRMB consists of two subsets: LLM-LLM pairs, where both chosen and rejected stories are generated by LLMs, and Human-LLM pairs, where the chosen story is human-written and the rejected one is LLM-generated. We evaluate the reward models on these two subsets and the results are shown in Figure 2. We find that existing models perform significantly worse on Human-LLM pairs, indicating a tendency to prefer LLM-generated stories. One possible reason is that LLMs favor LLM-generated content (Liu et al., 2023). Consequently, such reward models often fail to guide models toward producing human-like stories. In contrast, STORYREWARD performs better on Human-LLM pairs, demonstrating stronger alignment with human preferences. This advantage stems from our effective collection of human-written preference pairs. Therefore, we call on the community to leverage more human-written data in training and preference modeling to advance toward truly human-level story generation.

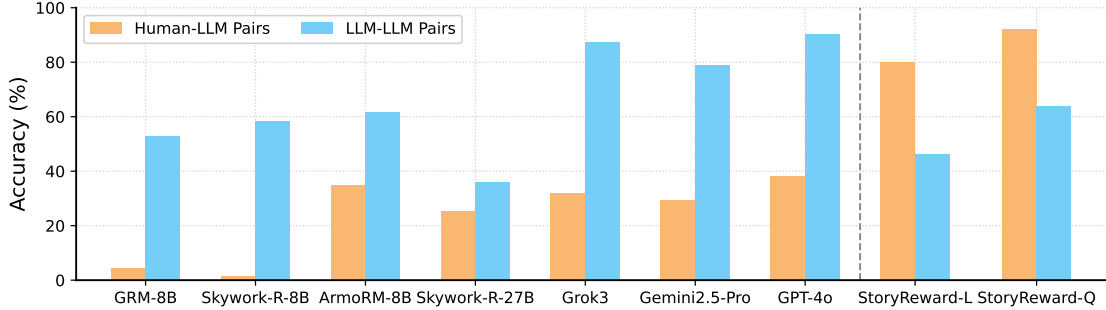


Figure 2: Accuracies (%) on LLM-LLM and Human-LLM pairs. LLM-LLM denotes all stories are generated by LLMs. Human-LLM denotes the chosen is human-written and the rejected is LLM-generated.

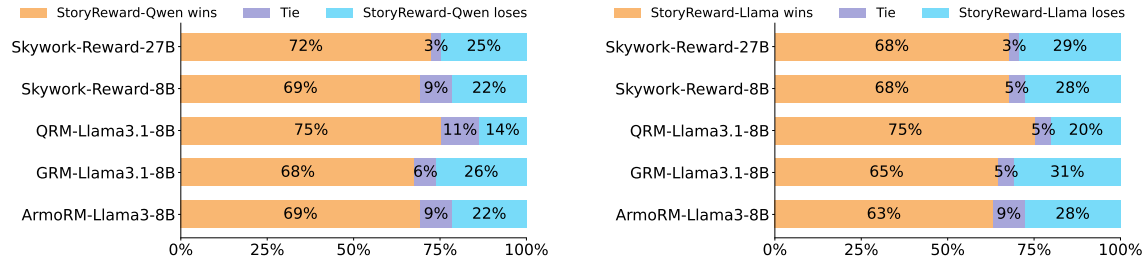


Figure 3: Head-to-head comparison results of BoN on the MoPS test set. Both STORYREWARD-QWEN and STORYREWARD-LLAMA show a dominant win rate. “Tie” means both models select the same story.

5 TEST-TIME SCALING WITH BEST-OF-N SAMPLING

An important application of reward models is test-time scaling, which allows the model to generate multiple responses and use reward models to select the best one, i.e., Best-of-N (BoN) sampling (Brown et al., 2024; Peng et al., 2025). In this section, we apply reward models in test-time scaling applications to evaluate the effectiveness of STORYREWARD. Specifically, we use the commonly used story premise dataset, the MoPS dataset (Ma et al., 2024), which contains approximately 100 premises. For each premise, we generate 16 stories using Llama-3.1-70B-Instruct (Meta, 2024a) with a sampling temperature of 1. Human annotators then rank the 16 stories according to preference, where two annotators perform rankings independently and any inconsistencies are judged by another senior annotator. We then adopt different reward models to select the best story among the 16 stories. We conduct a head-to-head evaluation, where the winning story is the one ranked higher in the human preference ordering.

Head-to-head results are shown in Figure 3. We observe that STORYREWARD substantially outperforms other reward models, with fewer than 30% of cases selecting an inferior story. This demonstrates that our models are more effective at identifying better stories during test-time scaling. Notably, our models are only of the 8B scale with low deployment and inference costs, which highlights their practicality for real-world test-time scaling. We also encourage the community to adopt our reward models for more fine-grained search, such as MCTS (Xie et al., 2024), enabling models to select even better stories.

6 RELATED WORK

6.1 STORY GENERATION

Story generation has advanced significantly with LLMs, prompting growing research into improving narrative quality. [Huot et al. \(2025\)](#) propose Agents’ Room, a generation framework inspired by narrative theory, that decomposes narrative writing into subtasks tackled by specialized agents. [Wang et al. \(2024b\)](#) propose Dynamic Hierarchical Outlining with Memory-Enhancement long-form story generation method, named DOME, to generate the long-form story with coherent content and plot. [Wang & Kreminski \(2024\)](#) propose to use a higher-level and more abstract symbolic specification of high-level story structure, implemented via answer set programming, to guide and diversify LLM-based story generation. [Lee et al. \(2025\)](#) propose WritingPath, a framework that uses explicit outlines to guide LLMs in generating goal-oriented, high-quality text, drawing inspiration from structured writing planning and reasoning paths, focusing on reflecting user intentions throughout the writing process. [Wan et al. \(2025b\)](#) introduces CogWriter to transform LLM constrained long-form text generation into a systematic cognitive writing paradigm. CogWriter consists of a Planning Agent that performs hierarchical planning to decompose the task and multiple Generation Agents that execute these plans in parallel. However, some studies argue that LLM-generated stories remain far from human-level quality ([Chakrabarty et al., 2024](#); [Tian et al., 2024](#); [Wang & Kreminski, 2025](#)), which we attribute in part to the lack of effective reward model guidance during training.

6.2 REWARD MODELING IN STORY GENERATION

LLMs have achieved remarkable breakthroughs in tasks with objectively verifiable solutions, such as mathematics and programming. These advances have been largely driven by Reinforcement Learning with Verifiable Rewards (RLVR), which leverages reference signals derived from rule-based verifiers to provide binary feedback (correct or incorrect). Such a paradigm is highly effective when tasks admit clear ground-truth answers. By contrast, writing tasks lack standardized solutions and are inherently subjective, making them far more challenging. LLM-generated texts are often verbose and hard to verify their quality. To address this challenge, [Jia et al. \(2025\)](#) introduces a new training paradigm designed for non-verifiable tasks like creative writing. Instead of relying on fixed human-written datasets, [Jia et al. \(2025\)](#) employs a dual-agent framework: a writing agent generates stories while a reviewing agent provides detailed, constructive feedback. This feedback serves as a rich, scalable reward signal, enabling iterative self-improvement without extensive human supervision. Although not specifically targeted at story generation, this framework represents an important step toward building reward models that capture the subtleties of open-ended creative tasks. Parallel to this, [Fein et al. \(2025\)](#) has emerged as the first standardized benchmark and paired dataset for preference modeling in creative writing. By curating data collected from Reddit, [Fein et al. \(2025\)](#) provides 43,827 human-labeled preference pairs, establishing a foundation for training and evaluating reward models.

7 CONCLUSION

In this paper, we systematically evaluate and train reward models for story generation. Specifically, we propose STORYRMB, the first benchmark for evaluating reward models on story preferences. We assess several representative reward models and LLM-as-judges. We find that current reward models poorly capture human story preferences and often favor LLM-generated over human-written stories. To address this gap, we introduce STORYREWARD, an advanced reward model for story preference, trained on a large-scale, high-quality dataset collected via our automated pipeline, which gathers human-written stories and real human preference signals, and applies techniques such as rewriting for diverse coverage. STORYREWARD achieves state-of-the-art performance on STORYRMB. We also adopt reward models in test-time scaling applications, using them for BoN searching and find STORYREWARD generally chooses better stories. We suggest that the research community develop enhanced story reward models to achieve truly human-level story generation.